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Non-Invasive Touch-Based Lithium Monitoring Using an Organohydrogel-Based Sensing Interface

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Lithium is a drug widely employed for the treatment of bipolar disorder owing to its high efficacy in mood management and suicide prevention. However, this efficacy is often undermined by misdosing and nonadherence, and diligent drug monitoring is required during treatment. Standard lithium monitoring involves invasive blood collections and laboratory analysis with low time granularity. Recent advances in sensor technology have enabled the development of personalized drug-monitoring devices that analyze biomarker information noninvasively. Herein, based on the fact that the analyte partition onto the fingertip with a high flux, a touch-based noninvasive monitoring modality for managing lithium pharmacotherapy is devised. The system is built based on a thin organohydrogel-mounted lithium ion-selective electrode (TOH-ISE). The TOH coating provides a stable environment for sensing. Through the utilization of a water/glycerol bi-solvent matrix, the gel exhibits dehydration-resist properties, rendering a controlled micro-environment for ISE conditioning, and subsequently minimizing signal drift. To illustrate the clinical application of the solution, the system is tested on a subject prescribed lithium. The system successfully detected the increase in circulating drug levels following medication intake. Collectively, the results indicate the devised solution is capable to facilitate lithium adherence monitoring and has broader potential for optimizing lithium pharmacotherapy.

1. Introduction

The lithium salt is widely prescribed as psychiatric medications for people with bipolar disorder.[1] Despite its demonstrated efficacy in mood stabilization, translating this effectiveness into improved patient outcomes in clinical practice remains challenging and unpredictable. First, lithium has a narrow therapeutic window (≈0.6 – 1.2 mM) with severe adverse effects if overdosed.[2] Second, due to the mental health of the specific patient population of interest and the medication itself (e.g., bipolar illness symptoms, drug's side effects), the adherence rate of lithium pharmacotherapy is low (≈60%).[3,4] Poor adherence can lead to poor treatment outcomes associated with an increased risk of relapse, rehospitalization, and suicide.[5] The standard procedure of lithium monitoring for precise dosing is restricted to centralized laboratories, which requires invasive blood collection and costly lab-based analysis with a prolonged turnaround

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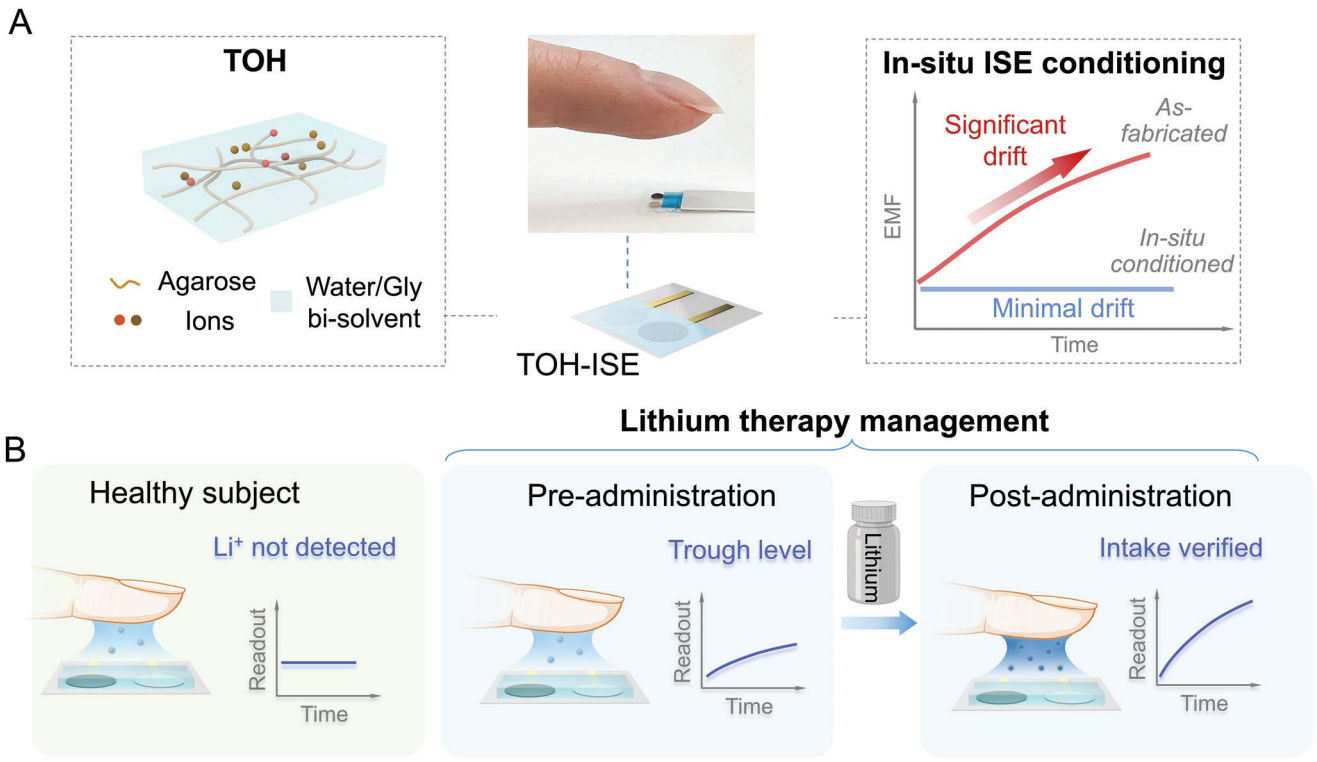


Figure 1. Touch-based lithium monitoring: design rationale and application. A) The schematic and the optical image of a TOH-ISE interface (center). The left panel highlights the composition and the structure of a TOH with an anti-dehydration property. The right schematic shows minimal sensor drift with the aid of in situ ISE conditioning. B) Application of the TOH-ISE interface for lithium therapy management.

time. Furthermore, there is currently no demonstrated method that directly monitors lithium adherence. The indirect monitoring solutions, such as pill counters, are incapable of verifying actual intake and accounting for the variable pharmacokinetics among subjects, which inherently lack specificity.

Wearable and mobile biochemical sensing technologies are suitable to overcome these limitations, because they can potentially be utilized at a large scale to monitor molecular-level information in a non-/minimally-invasive and real-time manner.[6–8] Particularly, recent studies have revealed that certain circulating biomarker molecules tend to accumulate on the surface of fingertips, primarily through natural perspiration, with a relatively high flux, which can be leveraged for user-friendly and practical biomonitoring by overcoming the challenges of low sample volume and physical/biochemical interference.[9–11]

Seeing an opportunity for decentralized lithium therapy management via biomonitoring, we developed a touch-based non-invasive lithium monitoring solution, which centers on a gel-coated lithium sensing interface to collect and analyze the partitioned lithium ions on fingertips in situ. A thin organohydrogel-coated lithium ion-selective electrode (TOH-ISE) was utilized to construct this interface, which features a uniquely developed TOH that simultaneously addresses stability challenges associated with the sensor and the sensing modality (**Figure 1**A). To elaborate, from the sensing interface aspect, previously demonstrated touch-based sensors mainly use a hydrogel interface for analyte sampling from the fingertip. However, the rapid hydrogel dehydration complicates the device's storage and usage.[12,13] From the ion-selective sensing aspect, prolonged ISE conditioning is usually required to render low-drift readouts,[6,14] which represents the main challenge limiting their lab-to-field translation. To overcome these challenges, we devised a TOH that was endowed with an anti-dehydration property by adopting a water-glycerol bi-solvent matrix.[15,16] When coupled with a lithium ISE, the TOH coating also function as a stable micro-environment to condition the ISE in situ. We thus eliminate the need for extensive ISE conditioning and thus enable a "plug-and-sense" operation.

The developed TOH-ISE was further augmented with a custom-developed ISE-specific signal interpretation framework—capable of extracting the lithium flux information from the touch-based readouts. The integrated lithium monitoring solution was then validated in both ex situ and in situ settings, by leveraging a custom-developed artificial fingertip setup and performing human subject studies, respectively. Collectively, the results show that the devised touch-based solution possess reliable lithium sensing capabilities, demonstrating its potential for monitoring lithium adherence and managing lithium pharmacotherapy (Figure 1B).

2. Result

Our designed touch-based lithium sensing interface consists of a thin gel (here, TOH)—which simultaneously serves as a bio-interface to sample the analyte flux from the skin surface and a medium for electrochemical analysis—and a lithium ISE to

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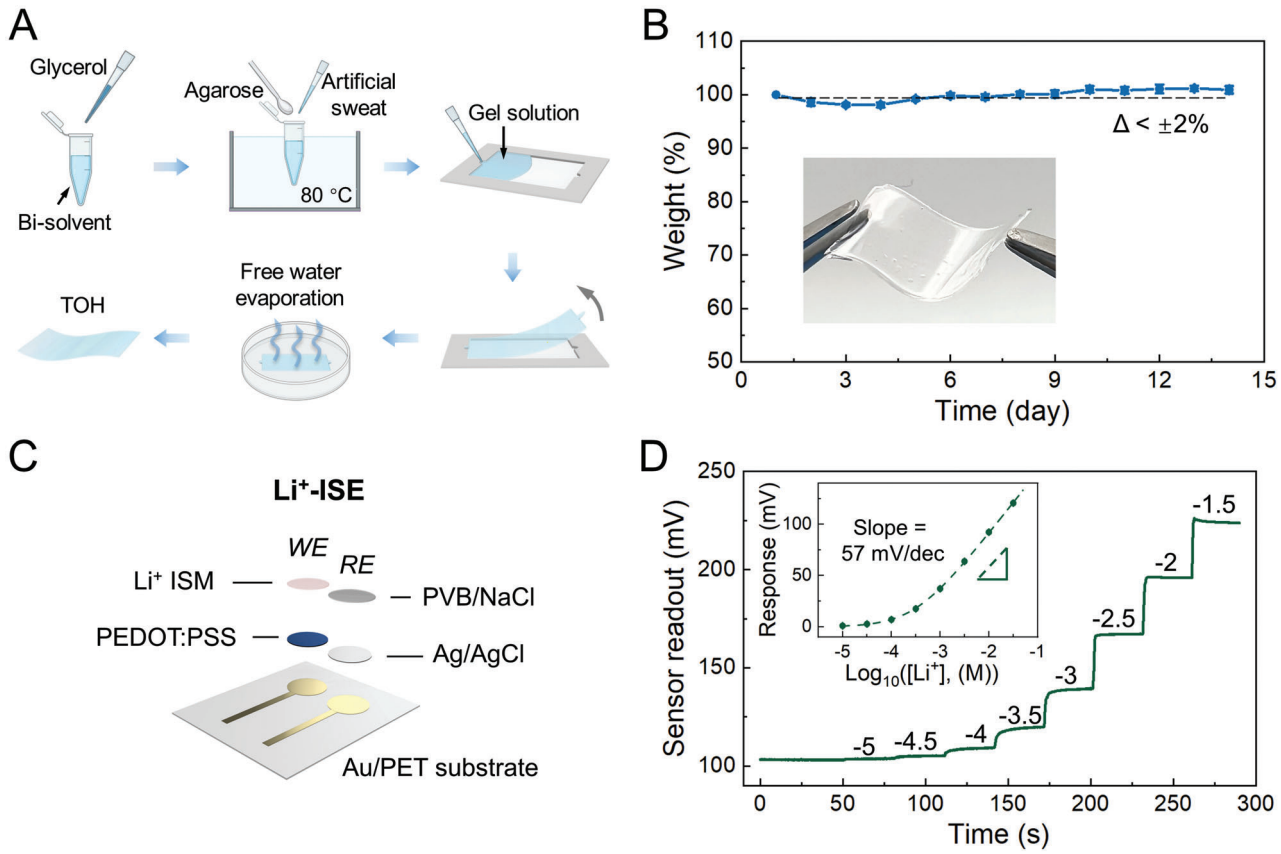


Figure 2. Characterization of individual components of the sensing interface. A) Schematics of the TOH fabrication workflow. B) Normalized TOH weight over 14 days. Error bars indicate standard deviation ($n = 10$). C) Exploded view of a Li^+ ISE (WE: working electrode; RE: reference electrode). D) Potentiometric readout of a representative Li^+ ISE with increasing Li^+ (lithium chloride, LiCl) concentration. Inset shows the corresponding calibration plot. Error bars indicate standard deviation of three sensors. The tests were performed in an electrochemical cell containing artificial sweat solution.

quantify sampled lithium ions. The lithium detection capability of the interface is thus determined by both the ISE (sensitivity/selectivity) and the gel (analyte transportation/dilution during sampling). Here, the two components were first individually developed and characterized.

The agarose-based TOH was fabricated using a molding technique (**Figure 2**A). Indispensable to rendering an antidehydration property, the gel solution was prepared in a bi-solvent matrix containing water and glycerol, where hydrogen bonds were formed between water, glycerol, and agarose molecules to "freeze" water molecules within the gel matrix.[17] The as-fabricated organohydrogels were then dried overnight to evaporate unbound free water molecules, resulting in a stabilized water-glycerol gel network (i.e., TOH). As shown in Figure 2B, the fabricated TOHs demonstrated negligible weight loss over a period of two weeks in an ambient environment, illustrating its superior anti-dehydration capability. Moreover, the TOHs have a small thickness of ≈60 µm, which facilitates rapid TOH-sensor response and amplifies the signal.[18]

The underlying lithium sensing interface was constructed using a solid-contact ISE (Figure 2C), which consists of: 1) an ion-selective membrane (ISM), serving as the recognition element for selective lithium chemisorption; 2) a poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS)-based ion-to-electron transducer, featuring a large redox capacitance to render low sensor drift; and 3) a gold electrode patterned on a flexible polyethylene terephthalate (PET) substrate. Here, the composition of the ISM cocktail was specifically selected toward optimal performance in a sweat-mimicking matrix (Figure S1, Supporting Information).

The developed sensor was first characterized in a standardized matrix. In a deionized water background, the sensor shows a Nernstian response to Li^+ within the range of 10^{-5}-$10^{-1.5}$ M (Figure S2A, Supporting Information), which is comparable to the previously reported Li^+-ISEs.[19,20] The intrinsic selectivity of the developed sensor was evaluated by testing the sensor in solutions containing either the interfering ion (e.g., Na^+, K^+, Ca^{2+}, Mg^{2+}) or Li^+. As shown in Figure S2B (Supporting Information), the potentials recorded with all the interfering ions were much lower than the signal introduced by Li^+. Selectivity coefficients were not defined because the sensor did not give a Nernstian response to interfering ions.[21] More informatively, considering that natural perspiration is the primary source of touch-based sensing, the developed sensor was further characterized in a custom-developed artificial sweat matrix,[22] which effectively mimics the ionic environment of sweat. The sensors' responses exhibited a linear range slope (m) of 57 mV/dec (Figure 2D). They can detect Li^+ concentration as low as 10 µM, which is substantially lower than the concentration of Li^+ in non-invasively retrievable biofluids (e.g., saliva: 0.5–3 mM; ~ double the corresponding

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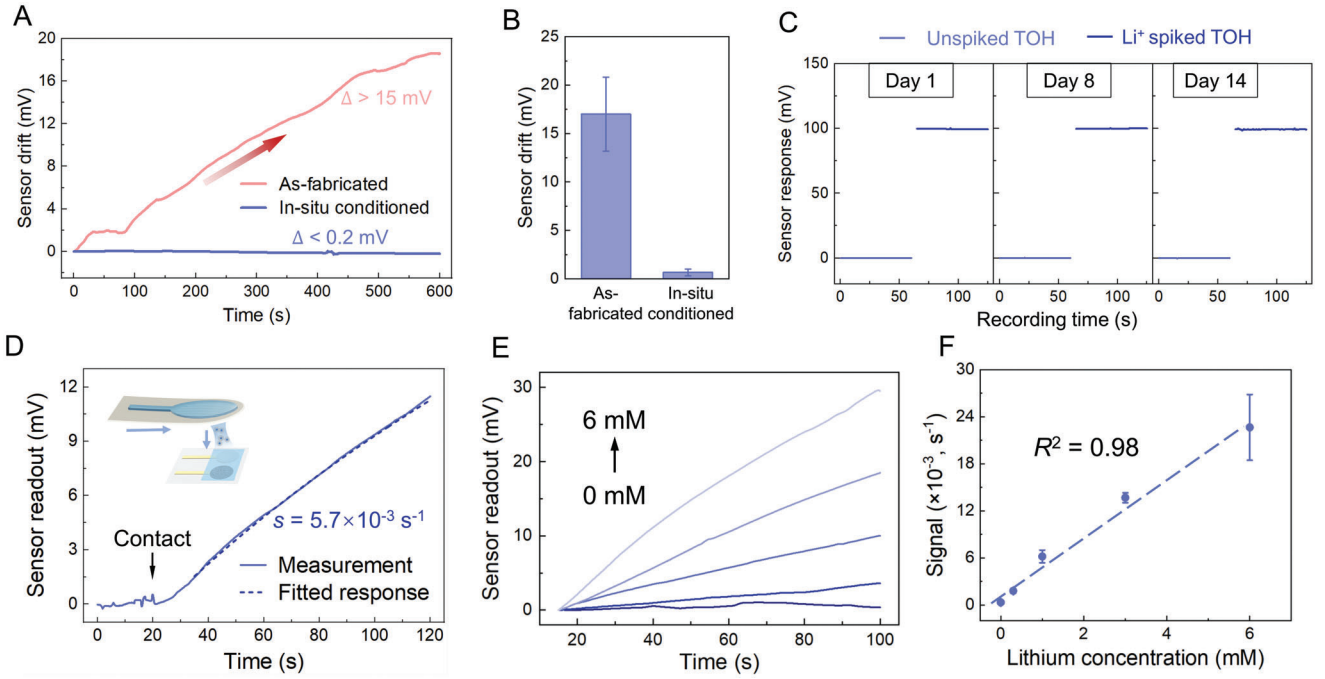


Figure 3. Ex situ TOH-ISE characterization. A) Real-time potentiometric readout drift of a representative TOH-ISE in its as-fabricated state (right after the coupling of TOH and ISE) and after overnight conditioning (in situ conditioned state). B) Corresponding sensing signal drift magnitude in the first 10 min of the test. Error standard deviation ($n = 5$, $p < 0.01$). C) Potentiometric response of a TOH-ISE interrogated on different days after fabrication. Lithium-spiked TOH (1 mM Li^+ in the prepared gel solution) was used to replace the TOH in order to test the response of the sensing interface. D) Potentiometric recording of a TOH-ISE upon contact with a microfluidic artificial fingertip (delivering artificial sweat solution with 1 mM Li^+). The dotted line shows the corresponding signal interpretation framework fitted readout. Inset shows the schematic of the testing setup. E) Potentiometric responses of a TOH-ISE to input fluid with various Li^+ concentrations (0, 0.3, 1, 3, 6 mM, all post contact with the artificial fingertip). F) Corresponding calibration curve. Error bars indicate standard deviation (three trials).

serum concentrations).[23,24] Furthermore, Figure S2C (Supporting Information) shows that the sensor's response to the variations of interfering ions was minimal as compared to that of Li^+, demonstrating good selectivity in the envisioned clinical application.

The individually-characterized TOH and lithium ISE were then coupled to form a TOH-ISE sensing interface, where the TOH serves as the sensing medium and ISE readout reflects the lithium level in the TOH.

We first characterized the TOH-ISE's readout stability. In conventional ISE sensing, to realize a low-drift readout, the sensor needs to be extensively conditioned in a sample-mimicking environment prior to the test, which is operationally complicated and time-consuming.[25] In our TOH-ISE configuration, the high stability of the TOH is leveraged to serve as a stable micro-environment to condition the ISE in situ. To demonstrate its utility, we compared the TOH-ISEs' readout drift in their as-fabricated state (i.e., right after the coupling of TOHs and ISEs) and in situ-conditioned state (i.e., after overnight storage in the coupled form). As shown in **Figure 3**A,B, the as-fabricated interface drifted significantly, while a small variation was observed in the in situ conditioned state. The small drift can be attributed to the equilibrium of the functional layers in the ISE (e.g., hydration, ion activity in the ISM, PEDOT:PSS redox state) during the conditioning process.[25] Leveraging this feature, the TOH-ISE interface effectively eliminates the time-consuming ISE conditioning step before use each time, rendering a "plug-and-sense" capability. To illustrate this capability, the developed TOH-ISE was interrogated on different days postfabrication. A lithium-spiked TOH was utilized to probe the interface's lithium response. As shown in Figure 3C, the interface manifested reproducible lithium responses for two weeks and the sensor readouts were stable for all the measurements.

To deploy TOH-ISE for touch-based sensing, the unique analyte transport behavior in touch-based sensing must be examined in relation to the TOH-ISE readout. Unlike the conventional sample-to-answer biochemical sensing scenarios, in touch-based sensing, the epidermal influx of the analyte into the gel-based interface (f) dynamically changes the analyte concentration/distribution in the gel. Furthermore, with a low analyte concentration, the non-linearity of the ISE response must be considered, which, together with the logarithmic nature of the ISE response, makes the signal interpretation non-trivial.

To develop an ISE-specific signal interpretation framework for touch-based lithium sensing, we start by referring to the Nicolsky–Eisenman equation, which empirically describes the ISE readout (EMF) in relation to the ion concentration levels in its surrounding environment [21] as shown in Equation (1):

$$EMF = \frac{RT}{z_I F} \log\left(a_I + \sum K_{IJ} a_J^{z_I/z_J}\right) + E_0 \quad (1)$$

Here a and z are the activity and the charge of the target ion (I; in our case Li^+) and non-target ion (J), respectively; K_{IJ} is the

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selectivity coefficient for each interfering ion; E_0 is the standard potential of EMF; and R, T, and F represent the universal gas constant, the absolute temperature, and the Faraday constant, respectively. For a touch-based lithium input, a_I gradually increases due to the epidermal analyte influx ($a_I = f \cdot t$), while the concentration variations of non-target ions have minimal influence on the readout. The latter part is achieved by optimizing the ionic composition of the TOH (Figure S3, Supporting Information). As a result, the dynamic sensor readout can be presented as Equation (2):

$$EMF\ (t) = \frac{RT}{z_I F}\ \log\left(f_I \cdot t + \sum K_{IJ} a_J^{\frac{z_I}{z_J}}\right) + E_0$$
$$= m \cdot \log\left(s \cdot t + 1\right) + E_0^{'} \quad (2)$$

Thus, the sensing signal (s, proportional to analyte flux) can be extracted by fitting the TOH-ISE readout into the equation above.

To validate the TOH-ISE interface and signal interpretation framework, we coupled the TOH-ISE with our previously developed microfluidic artificial fingertip setup, which mimics a skin surface that continuously delivers analytes onto the sensing interface.[18] As shown in Figure 3D, upon lithium introduction, the sensor readout gradually increased, whereas minimal signal change is observed in absence of lithium (Figure S4, Supporting Information), demonstrating that the TOH-ISE readout reflects the introduced lithium flux. To further quantify the readout, the introduced lithium concentrations were modulated to model the variation of circulating lithium levels. Figure 3E,F shows the real-time potentiometric readouts and the corresponding sensing signal (extracted via the interpretation framework above). The high-level of correlation ($R^2 = 0.98$) demonstrates the suitability of the devised approach for touch-based lithium analysis. Relevant to the touch-based sensing context, we also investigated the influence of the pressing force variability on the TOH-ISE readout. Figure S5, Supporting Information shows that mounting weights on the TOH-ISE did not cause sensor readout change, which is primarily due to the minimal electron motion involvement in the case of potentiometric sensing.

We further validated the performance of the developed TOH-ISE in an in situ lithium adherence monitoring study. To this end, a subject prescribed lithium-based medication was recruited. Touch-based sensing using the TOH-ISE interface was performed before and after the scheduled drug intake (900 mg lithium carbonate): ≈0–1 h pre-intake and ≈1–2 h post-intake. To demonstrate the physiological significance of the touch-based readouts, saliva samples were collected at the same time with touch-based sensing and subsequently analyzed using a standard colorimetric assay. The choice of saliva was motivated by lithium's reported saliva-blood correlation as well as the similar analyte partitioning pathway of sweat and saliva.[23] In a control experiment, touch-based sensing and salivary lithium analysis were also performed on three healthy subjects, where circulating lithium is expected to be negligible.

Figure 4A–C shows the representative raw and fitted TOH-ISE readouts from the three scenarios. From the patient inputs, a larger signal was obtained post-drug intake as compared to the pre-intake case, inferring an elevated circulating lithium level. On the other hand, a near-zero signal was recorded from the healthy subject. Figure 4D shows the collective TOH-ISE readouts and the corresponding saliva lithium levels. Similar trends are observed between both groups, supporting the potential utility of the touch-based non-invasive sensing for direct drug adherence monitoring applications.

3. Conclusion

In conclusion, we developed a touch-based lithium sensing interface, which consists of a thin organohydrogel and a lithium ISE. The devised sensing interface uniquely features a highly stable "plug-and-sense" operation, which is enabled by the gel's anti-dehydration property (provided by the bi-solvent matrix) and the in situ ISE conditioning (provided by the organohydrogel coupling). By coupling the sensing interface with an ISE-specific touch-based sensing signal interpretation framework, lithium influx monitoring was demonstrated using an ex situ setup as well as in a human subject study, illustrating its suitability for non-invasive lithium adherence monitoring.

To further improve the reliability of the system (i.e., signal-to-noise ratio) and the temporal granularity of the readout, future work is needed to address the confounding factors associated with touch-based sensing (e.g., variation of natural perspiration rate and temperature). To this end, axillary sensors like transdermal water loss sensor and temperature sensor can be integrated to normalize the readout.[6,26] Once established, a large-scale clinical investigation will be performed to correlate the sensor readouts with the circulating analyte concentration levels, where advanced machine learning-based algorithms can be utilized. Broadly, this technique can be further coupled with other novel medication adherence monitoring solutions,[27,28] where the non-invasive and low user intervention nature can be augmented with the advanced sample-to-answer biosensing capabilities (previously demonstrated with invasively accessed samples).[29,30] Collectively, we envisioned that the convergence of these efforts can ultimately enable decentralized pharmacotherapy management that can promote improved patient outcomes.

4. Experimental Section

Materials and Reagents: All the chemicals were purchased from Sigma–Aldrich, unless otherwise stated. Polyethylene terephthalate, 100 µm thick was purchased from MG Chemicals. Silver-silver chloride (Ag/AgCl) ink was obtained from Ercon Incorporated. Aqueous dispersion of PEDOT:PSS conductive polymer (Clevios PH 1000) was purchased from Heraeus Heraeus Precious Metals North America Daychem LLC. The artificial sweat solution was purchased from Pickering Laboratories, Inc. using a customized composition.[22]

Fabrication and Characterization of TOH: To fabricate TOH, 2 wt.% agarose and 3 wt.% artificial sweat solution (unless otherwise stated) were first mixed in a bi-solvent solution consisting of 80% deionized (DI) water and 20% glycerol. The mixture was homogenized in an 80°C water bath for 30 min. The resultant solution was then injected into vertically-assembled molds (fabricated following our previous protocol[31]). After sufficient cooling, a rectangular organohydrogel was formed. The gel was removed from the mold and dried overnight to allow for the evaporation of free water. The fabricated TOH was stored in an ambient environment if not used. To fabricate the lithium-spiked TOH, 1 mM lithium chloride (LiCl) was added into the gel solution before gelation. The thickness of TOH was measured to be 59 ± 10 µm (n = 6). The measurement was

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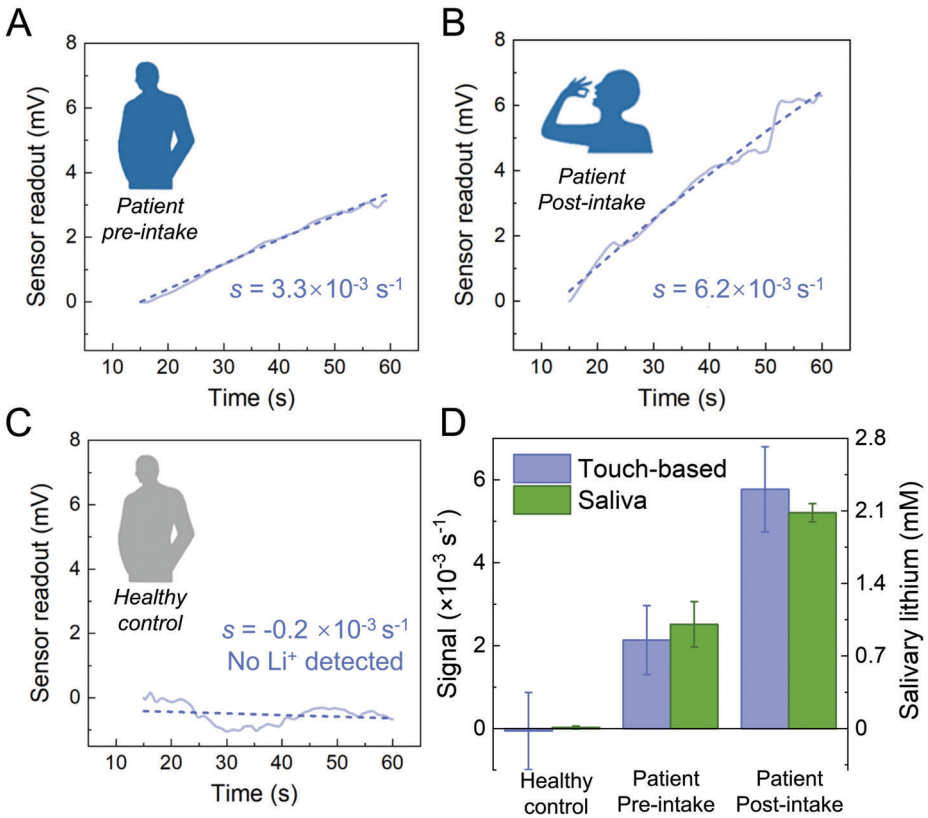


Figure 4. In situ TOH-ISE validation. A–C) Potentiometric TOH-ISE readouts and fitted responses on a patient before (A) and after (B) the scheduled daily lithium-based medication intake (900 mg lithium carbonate) as well as a healthy control subject (C). D) Healthy control's touch-based sensing signal and salivary lithium concentration versus the patient's measurements illustrating the negligible lithium concentrations in both measurements for the healthy control. Error bars indicate standard deviation. Healthy control: three subjects; Patient: three trials on the same subject. *p*-values <0.05.

performed under a microscope using a glass slide with a defined thickness as the reference.

Fabrication and Characterization of Lithium ISE: The lithium ISE was fabricated on a gold electrode, which was patterned on a PET substrate with the aid of a shadow mask (30 nm chromium/200 nm gold, electrode diameter: 2.4 mm). First, 2.5 µL PEDOT:PSS solution was dropped onto the gold electrode and then dried at room temperature (≈30 min). The acquired electrode was then baked at 60 °C for 30 min to facilitate PEDOT conjugation. For ISM coating, the PEDOT:PSS electrode was dipped into a lithium ISM cocktail for 10 dip cycles (2 s each) and the sensor was dried for 5 min between dips. To prepare the cocktail (cocktail 1), 3 mg Li Ionophore VI, 2 mg potassium tetrakis (4-chlorophenyl)borate (KTCPB), 84 mg polyvinyl chloride (PVC), and 203 µL 2-nitrophenyl octyl ether (NPOE) were dissolved into 3 mL tetrahydrofuran (THF).[32] To compare the performance of different ISM in our context at hand, a second widely-used lithium ISM cocktail (cocktail 2) was also prepared. This cocktail contained 4.5 mg Li Ionophore VI, 1.5 mg KTCPB, 4.5 mg trioctylphosphine oxide (TOPO), 84 mg PVC, and 198 µL NPOE in 3 mL THF.[20] The fabricated ISEs were conditioned in a 10 mM LiCl artificial sweat solution for 2 h before use. Standalone ISE characterizations were performed using a standard electrochemical cell (vs aqueous Ag/AgCl reference electrode, CH Instruments, Inc.,). All the electrochemical characterizations were performed using CHI660E or CHI1040C potentiostat (CH Instruments, Inc.,).

To fabricate the reference electrode, Ag/AgCl ink was first dropped onto the same gold electrode for ISE construction and dried at 65 °C for 30 min. Then a polyvinyl butyral/sodium chloride (PVB/NaCl) solution was dropped onto the Ag/AgCl electrode 3 times (4 µL per time; 10 min interval between dropping) to eliminate the interference of Cl^- concentration variation. The solution was prepared by dissolving 79 mg PVB and 50 mg NaCl into 1 mL methanol. The fabricated/conditioned lithium ISE and reference electrode were paired and taped on a PET substrate to form a full sensor.

Ex situ TOH-ISE Characterizations: The TOH was mounted onto a lithium sensor to form a TOH-ISE unit. The TOH-ISE was stored in an ambient environment if not used. The storage of the coupled unit also serves for in situ ISE conditioning. The unit was conditioned for at least overnight before any testing if otherwise stated.

The TOH-ISE's response to a lithium flux input was characterized by coupling with a microfluidic artificial fingertip setup that was fabricated following our previous work.[18] In the experiment, the potentiometric readout of the TOH-ISE was recorded continuously. After a stable baseline was obtained, the TOH-ISE was mounted onto the artificial fingertip, which delivered input fluid with different lithium concentration levels at a constant flow rate (320 nL/min/cm^2). Potentiometric readout between 15–100 s post artificial fingertip contact was fitted into the described signal interpretation model (Equation (2) in the main text) to extract the flux-related signals (MATLAB). In the fitting process, we used the experimentally determined value of the slope m (57 mV/dec, from the standalone ISE characterization) to avoid overfitting. The starting time point (15 s) was experimentally determined to account for the analyte diffusion time and the contact-induced disturbance. A new TOH was used for each measurement. To characterize the effect of pressing force, weights were used to emulate force exertion with different strengths.

In situ TOH-ISE Characterization: One subject under lithium-based pharmacotherapy (900 mg lithium carbonate per day) and three healthy subjects (without lithium-based medication history) were recruited in the study. For each touch-based entry, the subjects were instructed to wash their index or middle fingertips with DI water before testing. The acquired

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potentiometric readouts were processed using the same method as the ex-situ experiments to extract the sensing signal. Prior to collecting their saliva, the subjects were directed to rinse their mouth with cold water. A Saliva Collection Aid (Salimetrics) was utilized to directly collect the samples through salivation. The salivary lithium was quantified using a colorimetric lithium assay kit (Abcam ab235613) and Nanodrop One (Thermo Fisher).

Statistical analysis: All data were analyzed using MATLAB 2020a (including Curve Fitting Toolbox 3.5.11). Data visualizations were made by Origin 2020. *P*-values were calculated based on *t*-test.

Institutional Review Board Approval for Human Subject Testing: The conducted human subject experiments were performed in compliance with the protocols that have been approved by the Institutional Review Board at the University of California, Los Angeles (IRB No. 17–000170). All subjects gave written informed consent before participation in the study.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

S.E., S.L., and J.Z. are inventors on a provisional patent application (U.S. Provisional Patent Application No. 63/399,615) on the presented technology.

Author Contributions

S.L., and J.Z. contributed equally to this work. S.L., J.Z., and S.E. conceived the idea and designed the experiments. S.L. and J.Z. led the experiments (with assistance from J.Y., T.-Y.W., X.C., Y.Z., B.W., J.T., S.P., A.S., S.S, C.M., and S.E.); J.Y., T.-Y.W., X.C., Y.Z., B.W., J.T, and S.E. contributed to data analysis and interpretation; S.L., J.Z., and S.E. drafted the manuscript and all authors provided feedback. S.E. supervised the study.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

drug monitoring, personalized dosing, bipolar disorder, ion-selective sensors, analytical frameworks

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